



Chapter 2

Data Summary and Presentation





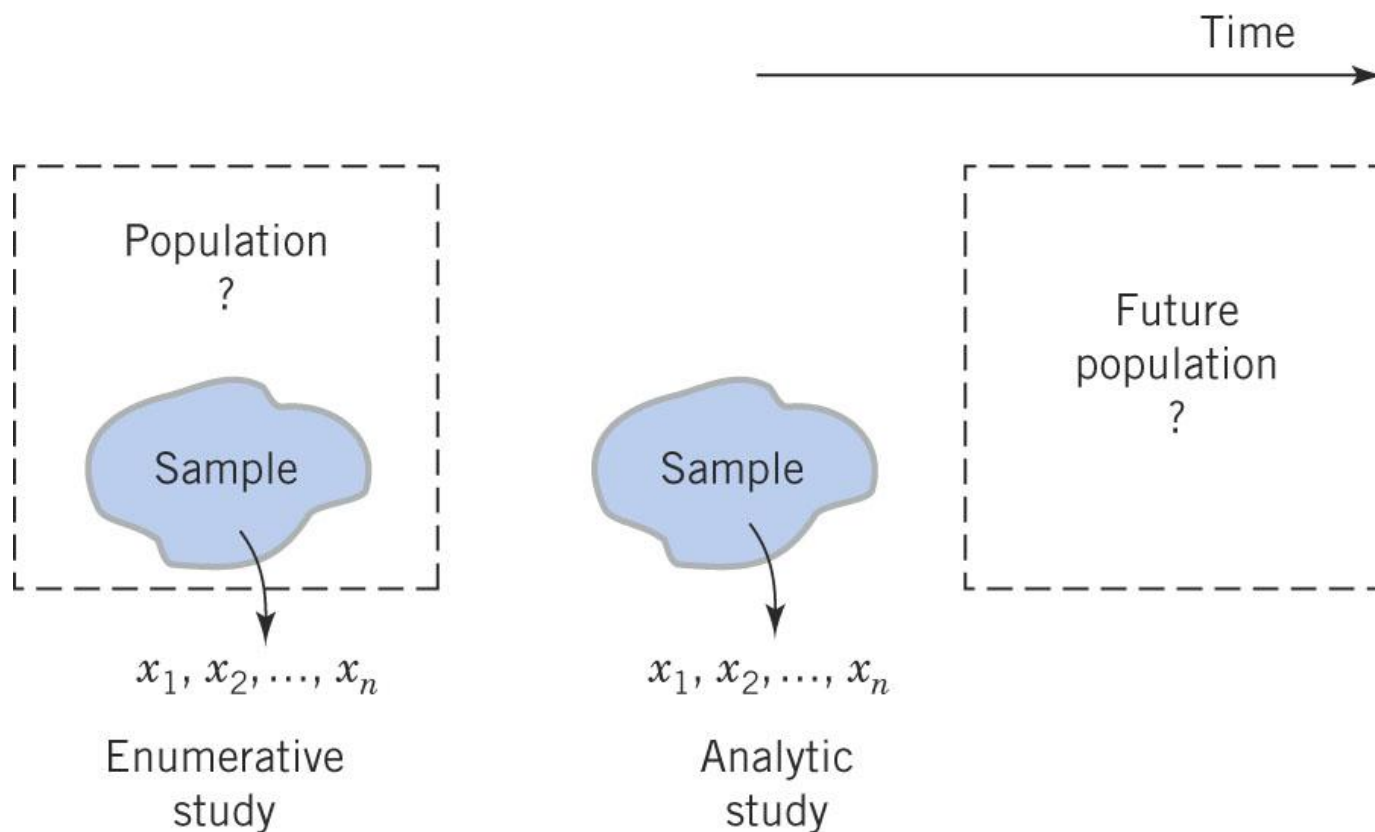
Data Summary

- ♣ Mean
- ♣ Variance and Standard Deviation
- ♣ Median and Mode
- ♣ Percentile and Quartile
- ♣ Range and Inter Quartile Range





Enumerative Versus Analytic Study





Mean

♣ Sample Mean

- ◆ If the n observations in a sample are denoted by $x_1, x_2, \dots, x_n$ the **sample mean** is

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

♣ Population Mean

- ◆ For a finite population with N measurements, the mean is

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

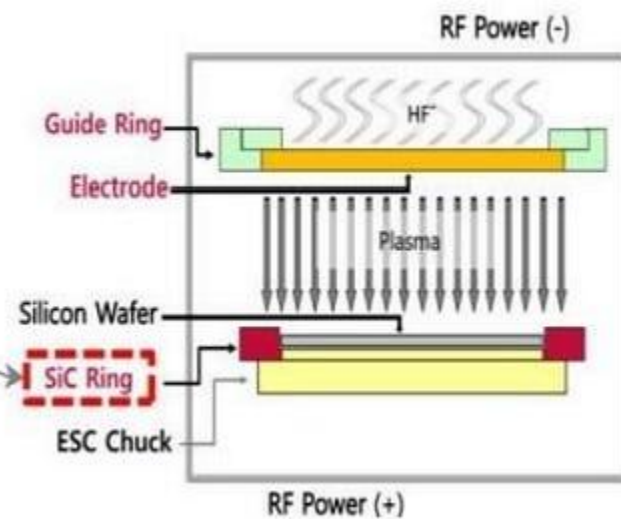
- ♣ The **sample mean** is a reasonable estimate of the **population mean**.

O-rings Example



- ❖ Suppose that an engineer is developing a rubber compound for use in **O-rings**.
- ❖ The O-rings are to be employed as seals in plasma etching tools used in the semiconductor industry, so their **resistance to acids and other corrosive substances** is an important characteristic.
- ❖ The engineer uses the standard rubber compound to produce **eight O-rings** in a development laboratory and measures the **tensile strength** of each specimen after immersion in a nitric acid solution at 30°C for 25 minutes.

SiC Focus Ring Shape





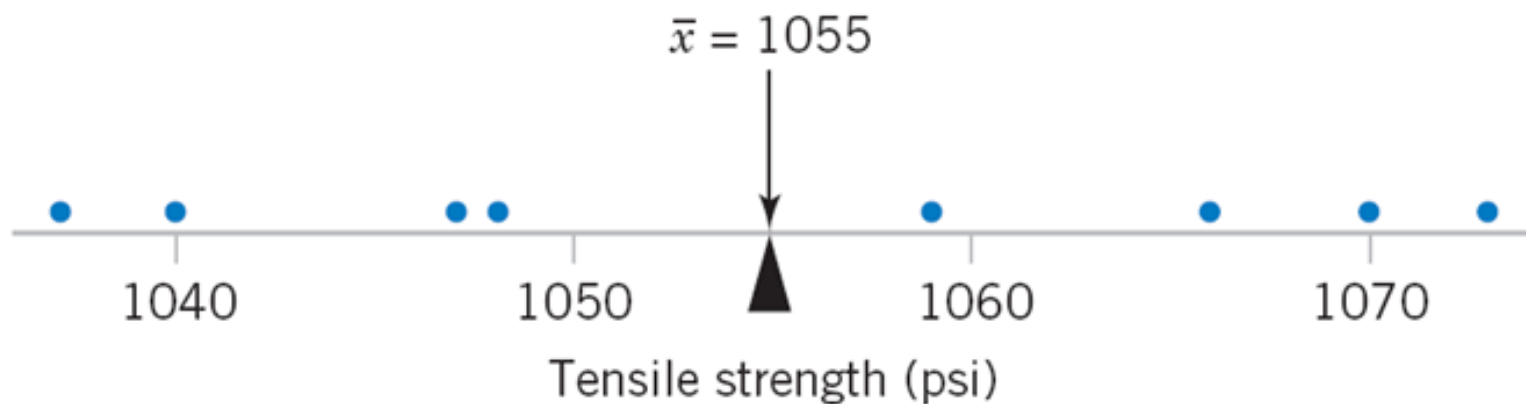
O-rings Example (Cont.)

- ♣ The tensile strengths (in psi) of the eight O-rings are 1037, 1047, 1070, 1066, 1059, 1073, 1048, and 1040.
- ♣ The sample mean strength (psi) for the eight observations on strength is

$$\begin{aligned}\bar{x} &= \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n} \\ &= \frac{1037 + 1047 + \dots + 1040}{8} = \frac{8440}{8} \\ &= 1055.0 \text{ psi}\end{aligned}$$



O-rings Example (Cont.)



Dot diagram of O-rings tensile strength. The sample mean is shown as a balance point for a system of weights.



Variance and Standard Deviation

♣ Sample Variance and Sample Standard Deviation

- ◆ If the n observations in a sample are denoted by $x_1, x_2, \dots, x_n$ the **sample variance** is

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

- ◆ The **sample standard deviation**, S , is the positive square root of the sample variance.



Variance (Cont.)

♣ Population Variance

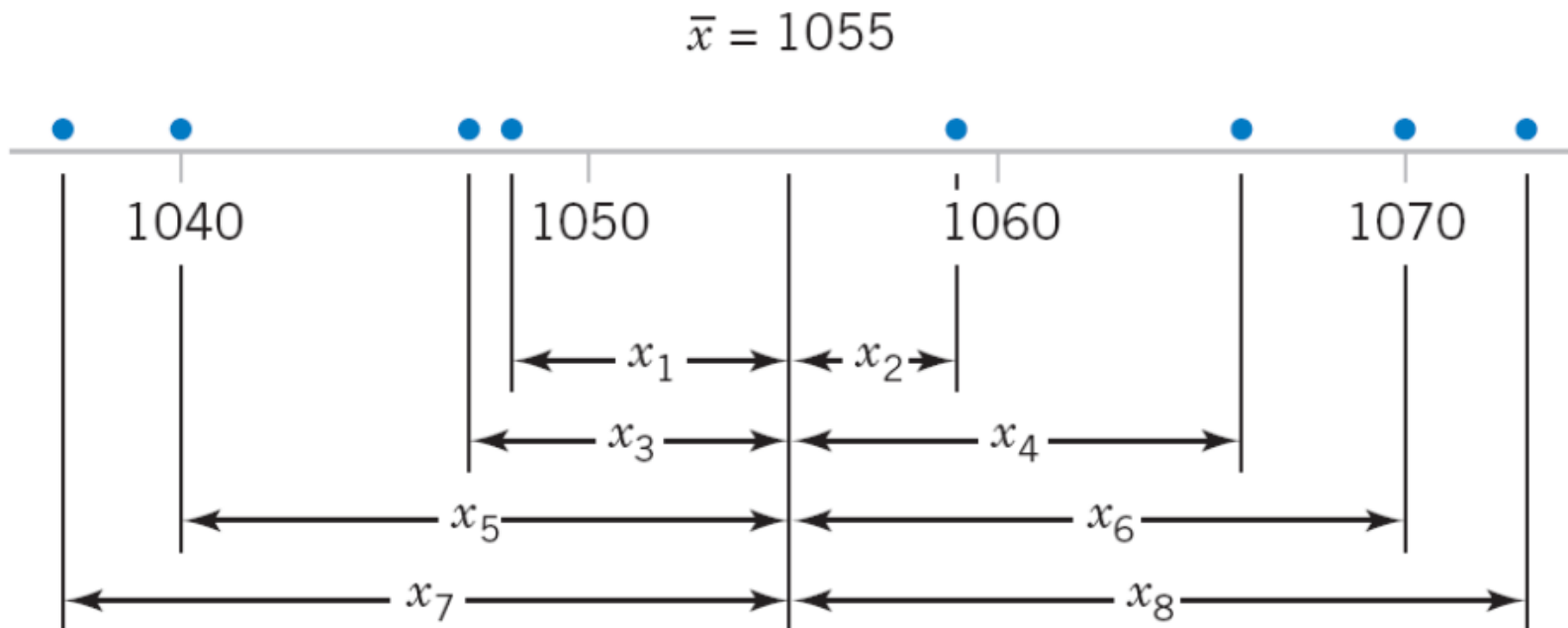
- ◆ When the population is finite and consists of N values, we may define the population variance as

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

- ♣ The **sample variance** is a reasonable estimate of the population variance.



O-rings Example (Cont.)



How the sample variance measures variability through
the deviation $x_i - \bar{x}$



O-rings Example (Cont.)

i	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	1048	-7	49
2	1059	4	16
3	1047	-8	64
4	1066	11	121
5	1040	-15	225
6	1070	15	225
7	1037	-18	324
8	<u>1073</u>	<u>18</u>	<u>324</u>
	8440	0.0	1348

Calculation of Terms for the Sample Variance
and Sample Standard Deviation



O-rings Example (Cont.)

♣ The sample variance is

$$s^2 = \frac{1348}{8-1} = \frac{1348}{7} = 192.57 \text{ psi}^2$$

♣ The sample standard deviation is

$$s = \sqrt{192.57} = 13.9 \text{ psi}$$



Variance

♣ Computational formula for s^2

$$s^2 = \frac{\sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n}}{n - 1}$$

♣ Why?

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

$$s^2 = \frac{\sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n}}{n - 1}$$



Median and Mode

- ♣ Mean, median, and mode are three kinds of "averages".
- ♣ The mean is the "average" you're used to, where you add up all the numbers and then divide by the number of numbers.
- ♣ The median is the "middle" value in the list of numbers. To find the median, your numbers have to be listed in numerical order, so you may have to rewrite your list first.
- ♣ The mode is the value that occurs most often. If no number is repeated, then there is no mode for the list.



Percentile and Quartile

♣ Percentiles

The K-th Percentile is that number such that **K% of all data values are less** and **(100 - K) % are larger** than it. For example, 80% of the data values lie below the 80th percentile.

♣ To find the K-th Percentile:

1. Sort all observations in ascending order
2. Compute the position $L = (K/100)^* (n+1)$, where n is the total number of observations.
3. Calculate the value at that position is the K-th percentile.



Percentile and Quartile (Cont.)

♣ Quartiles

Quartiles are values that divide a part of a data table into **four groups** containing an approximately equal number of observations. The total of 100% is split into four equal parts: 25%, 50%, 75% and 100%.

- ◆ The first quartile (or **lower quartile**), Q_1 , is the same thing as the twenty-fifth percentile.
- ◆ The third quartile (or **upper quartile**), Q_3 , is the 75 percentile.
- ◆ The second quartile (Q_2) is the median.



Range and Inter Quartile Range

♣ The **inter quartile range**, IQR, is defined as $Q3 - Q1$.

♦ **Exercise3:** 1, 3, 3, 4, 5, 6, 6, 7, 8, 8

♦ **IQR = ???**

♣ The **range** is just the difference between the largest and smallest values.

♦ **Range = ???**



Data Presentation

- ♣ Stem-and-Leaf Diagram
- ♣ Histograms
- ♣ Box Plot
- ♣ Time Series Plots
- ♣ Multivariate Data



Example

Compressive Strength of 80 Aluminum-Lithium Alloy Specimens



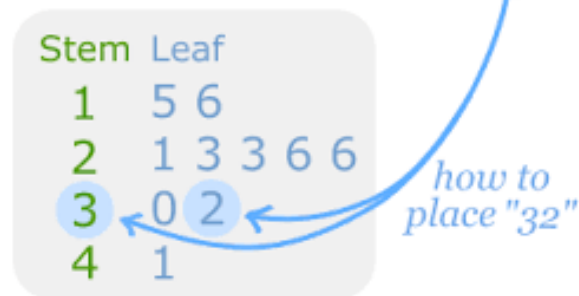
105	221	183	186	121	181	180	143
97	154	153	174	120	168	167	141
245	228	174	199	181	158	176	110
163	131	154	115	160	208	158	133
207	180	190	193	194	133	156	123
134	178	76	167	184	135	229	146
218	157	101	171	165	172	158	169
199	151	142	163	145	171	148	158
160	175	149	87	160	237	150	135
196	201	200	176	150	170	118	149



Stem-and-Leaf Diagram

- ♣ Stem-and-leaf diagram is a good way to obtain an information visual display of a data set $x_1, x_2, \dots, x_n$, where each number x_i consists of **at least two digits**.

15, 16, 21, 23, 23, 26, 26, 30, 32, 41





Steps for Constructing a Stem-and-Leaf Diagram

1. Divide each number x_i into two parts:
 - ◆ a **stem** - consisting of one or more of the leading digits, and
 - ◆ a **leaf** - consisting of the remaining digit.
2. List the stem values in a vertical column.
3. Record the leaf for each observation beside its stem.
4. Write the units for stems and leaves on the display.



Stem-and-Leaf Diagram
Example

Stem	Leaf	Frequency
7	6	1
8	7	1
9	7	1
10	5 1	2
11	5 8 0	3
12	1 0 3	3
13	4 1 3 5 3 5	6
14	2 9 5 8 3 1 6 9	8
15	4 7 1 3 4 0 8 8 6 8 0 8	12
16	3 0 7 3 0 5 0 8 7 9	10
17	8 5 4 4 1 6 2 1 0 6	10
18	0 3 6 1 4 1 0	7
19	9 6 0 9 3 4	6
20	7 1 0 8	4
21	8	1
22	1 8 9	3
23	7	1
24	5	1



Ordered Stem-and-Leaf Diagram

- ♣ The ordered stem-and-leaf diagram display makes it easy to find data feature such as percentiles, quartiles, and the median
- ♣ **Median** is a measure of central tendency that divides the data into two equal parts.
- ♣ **Range** = Max-Min
- ♣ When an ordered set of data is divided into four equal parts, the division points are called **quartiles**.
- ♣ **Interquartile range (IQR)** = $Q3 - Q1$
- ♣ **K Percentile** is a data value that approximately K% of the observation are at or below this value.



Ordered Stem-and-Leaf Diagram

	Stem	Leaf
1	7	6
2	8	7
3	9	7
5	10	1 5
8	11	0 5 8
11	12	0 1 3
17	13	1 3 3 4 5 5
25	14	1 2 3 5 6 8 9 9
37	15	0 0 1 3 4 4 6 7 8 8 8 8
(10)	16	0 0 0 3 3 5 7 7 8 9
33	17	0 1 1 2 4 4 5 6 6 8
23	18	0 0 1 1 3 4 6
16	19	0 3 4 6 9 9
10	20	0 1 7 8
6	21	8
5	22	1 8 9
2	23	7
1	24	5



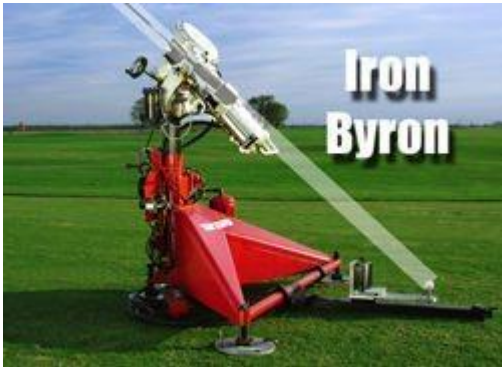
Histograms

- ♣ A **histogram** is a more compact summary of data than a stem-and-leaf diagram.
- ♣ To construct a histogram for continuous data, we must divide the range of the data into intervals, which are usually called **class intervals**, **cells**, or **bins**.
- ♣ If possible, the bins should be of equal width to enhance the visual information in the histogram.

Example: Golf Ball Distance

- ♣ The United States Golf Association tests golf balls to ensure that they conform to the rules of golf.
- ♣ The overall distance test is conducted by hitting ball by a mechanical device nicknamed *Iron Byron*.
- ♣ The distances by **hitting 100 golf balls** are recorded.
- ♣ Because the dataset contains 100 observations, we suspect that about $\sqrt{100} = 10$ bins will provide a satisfactory histogram.





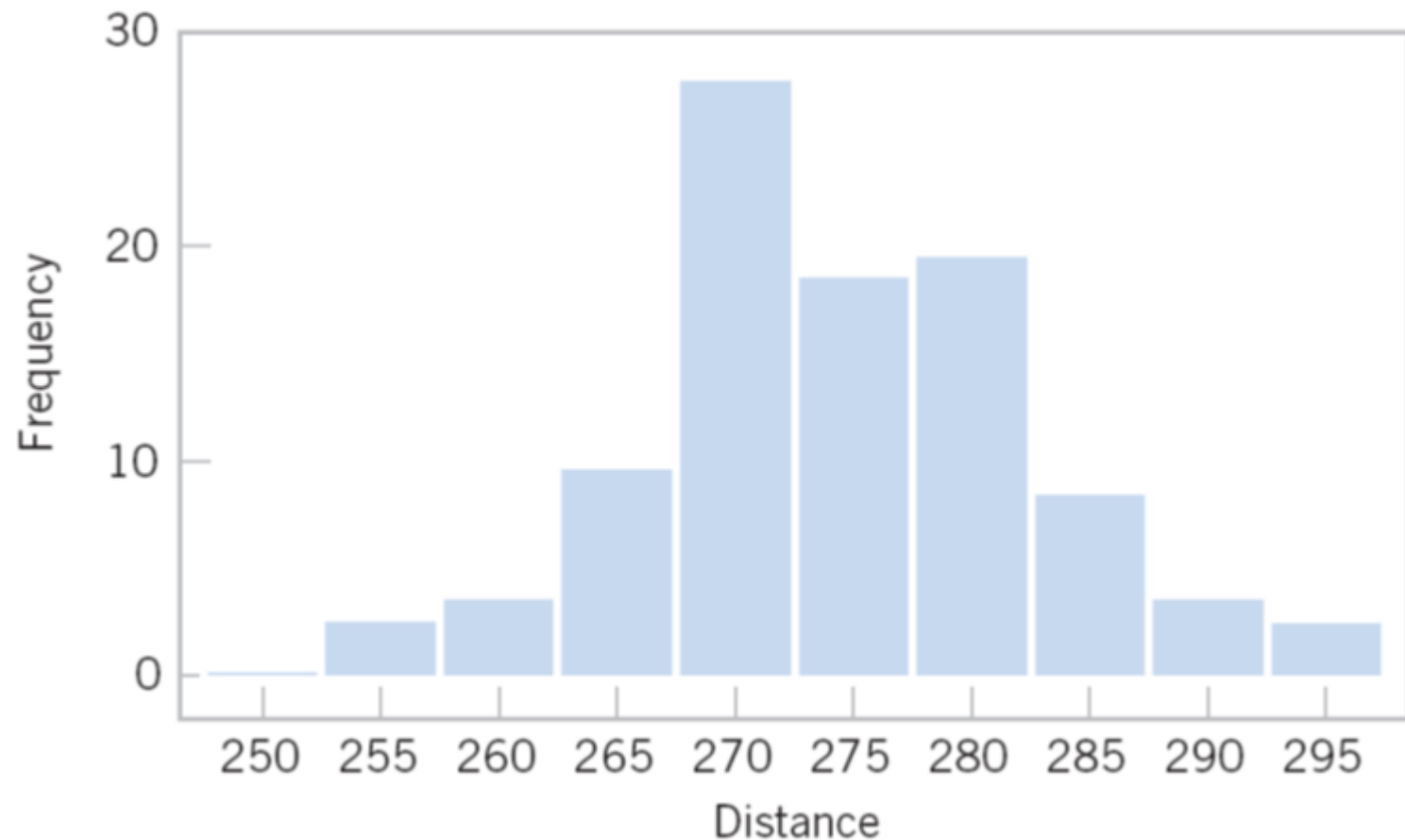


Golf Ball Distance (Cont.)

291.5	274.4	290.2	276.4	272.0	268.7	281.6	281.6	276.3	285.9
269.6	266.6	283.6	269.6	277.8	287.8	267.6	292.6	273.4	284.4
270.7	274.0	285.2	275.5	272.1	261.3	274.0	279.3	281.0	293.1
277.5	278.0	272.5	271.7	280.8	265.6	260.1	272.5	281.3	263.0
279.0	267.3	283.5	271.2	268.5	277.1	266.2	266.4	271.5	280.3
267.8	272.1	269.7	278.5	277.3	280.5	270.8	267.7	255.1	276.4
283.7	281.7	282.2	274.1	264.5	281.0	273.2	274.4	281.6	273.7
271.0	271.5	289.7	271.1	256.9	274.5	286.2	273.9	268.5	262.6
261.9	258.9	293.2	267.1	255.0	269.7	281.9	269.6	279.8	269.9
282.6	270.0	265.2	277.7	275.5	272.2	270.0	271.0	284.3	268.4



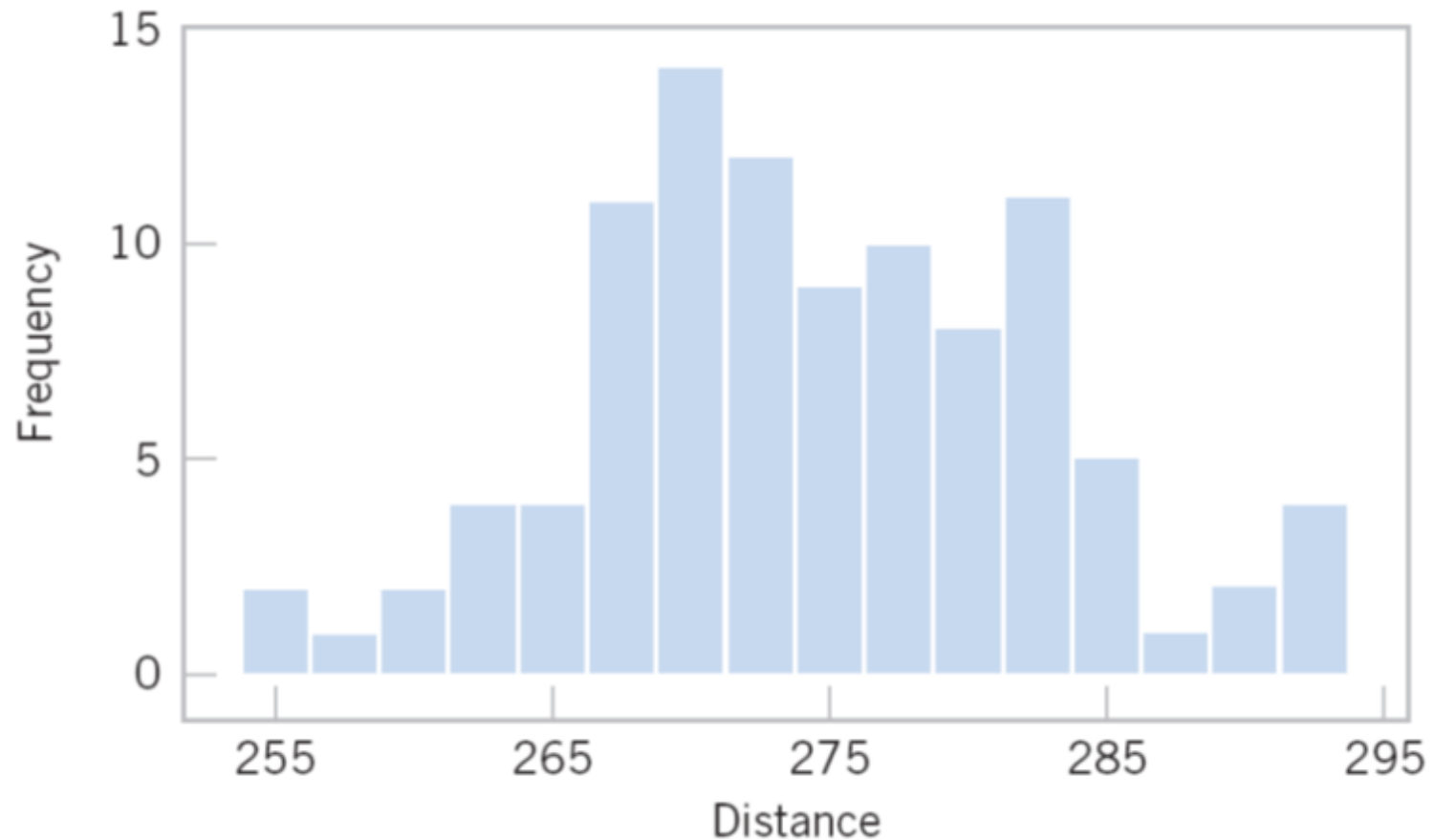
Golf Ball Distance (Cont.)



Histogram with 10 bins for the golf ball distance data.



Golf Ball Distance (Cont.)



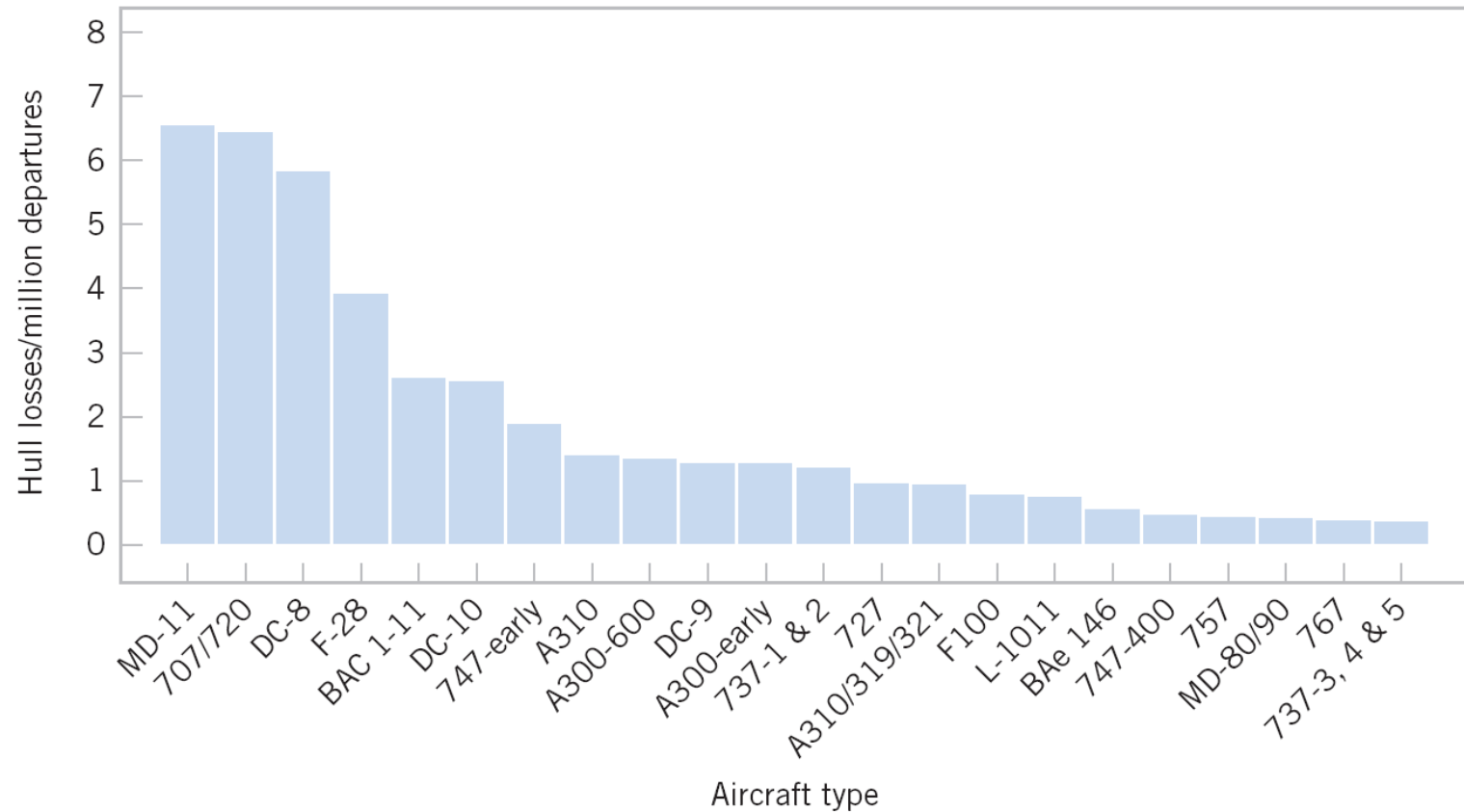
Histogram with 16 bins for the golf ball distance data.

Pareto Chart



- ♣ An important variation of the histogram is the **Pareto chart**.
- ♣ This chart is widely used in quality and process improvement studies where the data usually represent different types of defects, failure modes, or other **categories** of interest to the analyst.
- ♣ The categories are ordered so that the category with the largest number of frequencies is on the left, followed by the category with the second largest number of frequencies, and so forth.
- ♣ **Ordered by the frequency.**

Pareto Chart (Cont.)



Pareto chart for the aircraft accident data

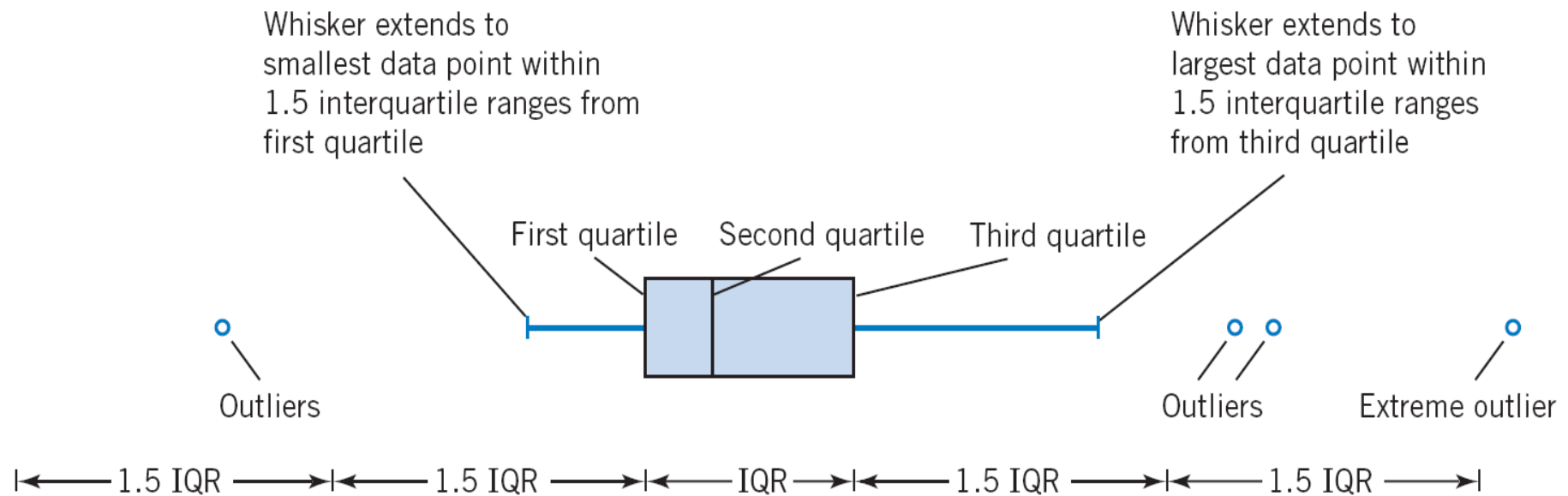


Box Plots

- ♣ The **box plot** is a graphical display that simultaneously describes **several important features** of a data set, such as center, spread, departure from symmetry, and identification of observations that lie unusually far from the bulk of the data.
- ♣ Three Other Components:
 - ◆ Whisker
 - ◆ Outlier
 - ◆ Extreme outlier

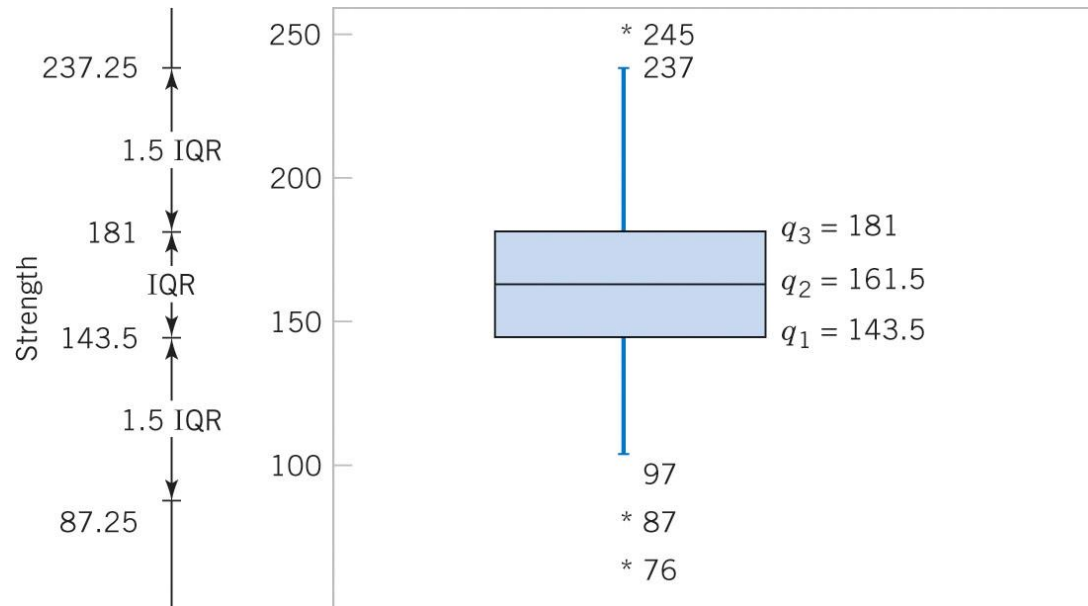


Box Plots (Cont.)



Description of a box plot

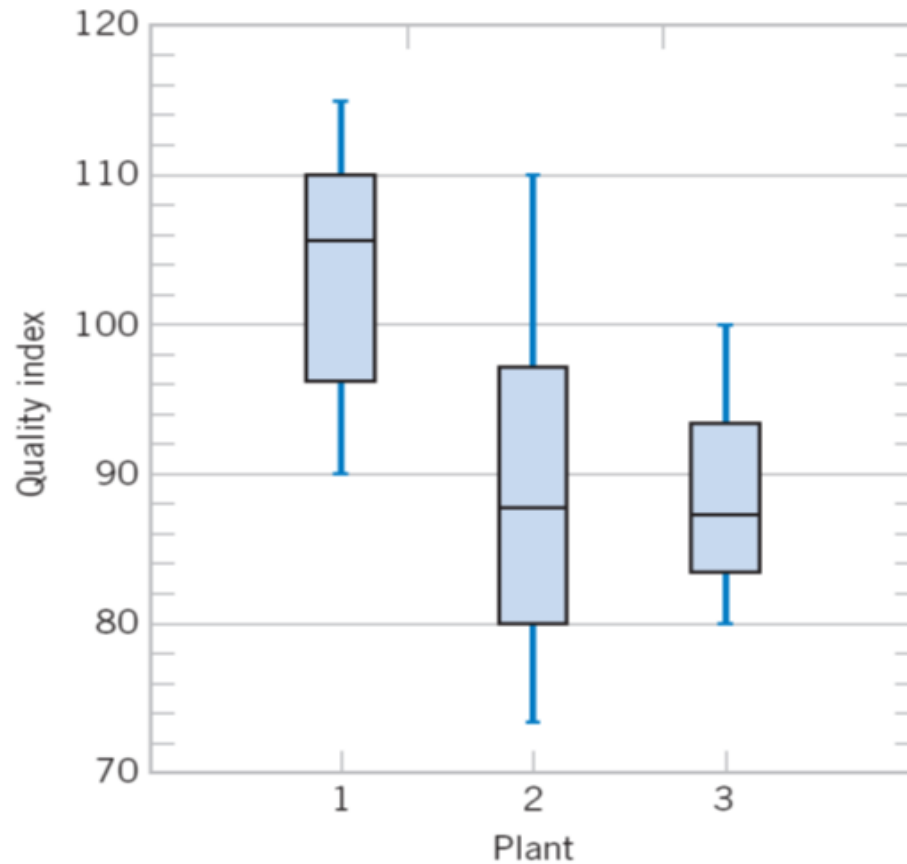
Box Plots (Cont.)



- Box plot of compressive strength of 80 Aluminium-Lithium alloy specimens.
- Comment: Box plot may be shown vertically or horizontally, data reveals three outliers and no extreme outliers.
- Lower outlier limit is: $143.5 - 1.5^*(181.0 - 143.5) = 87.25$.



Box Plots



Comparative box plots of a quality index at three plants

Time Series Plots

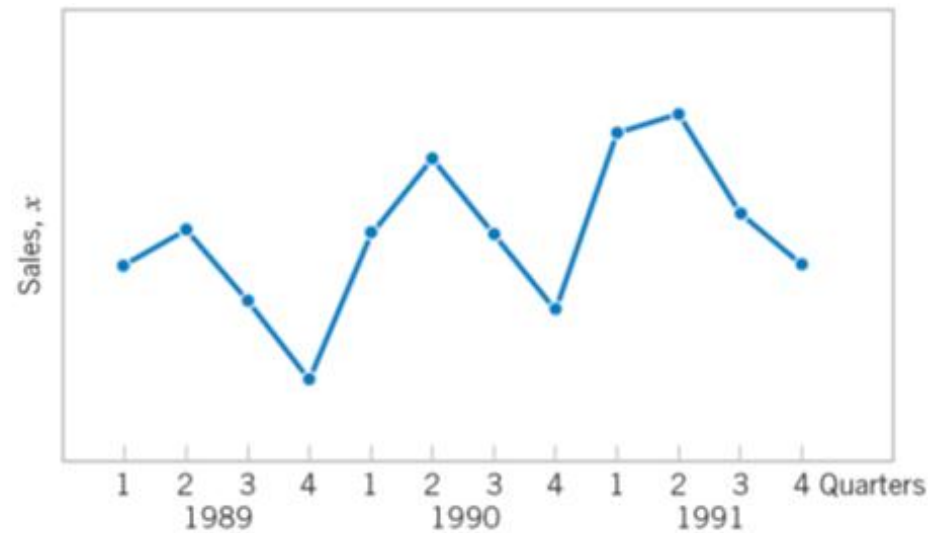
- ♣ A **time series** or **time sequence** is a data set in which the observations are **recorded in the order** in which they occur.
- ♣ A **time series plot** is a graph in which the vertical axis denotes the observed value of the variable and the horizontal axis denotes the time (which could be minutes, days, years, etc.).
- ♣ When measurements are plotted as a time series, we often see
 - ◆ trends,
 - ◆ cycles, or
 - ◆ other broad features of the data



Time Series Plots (Cont.)



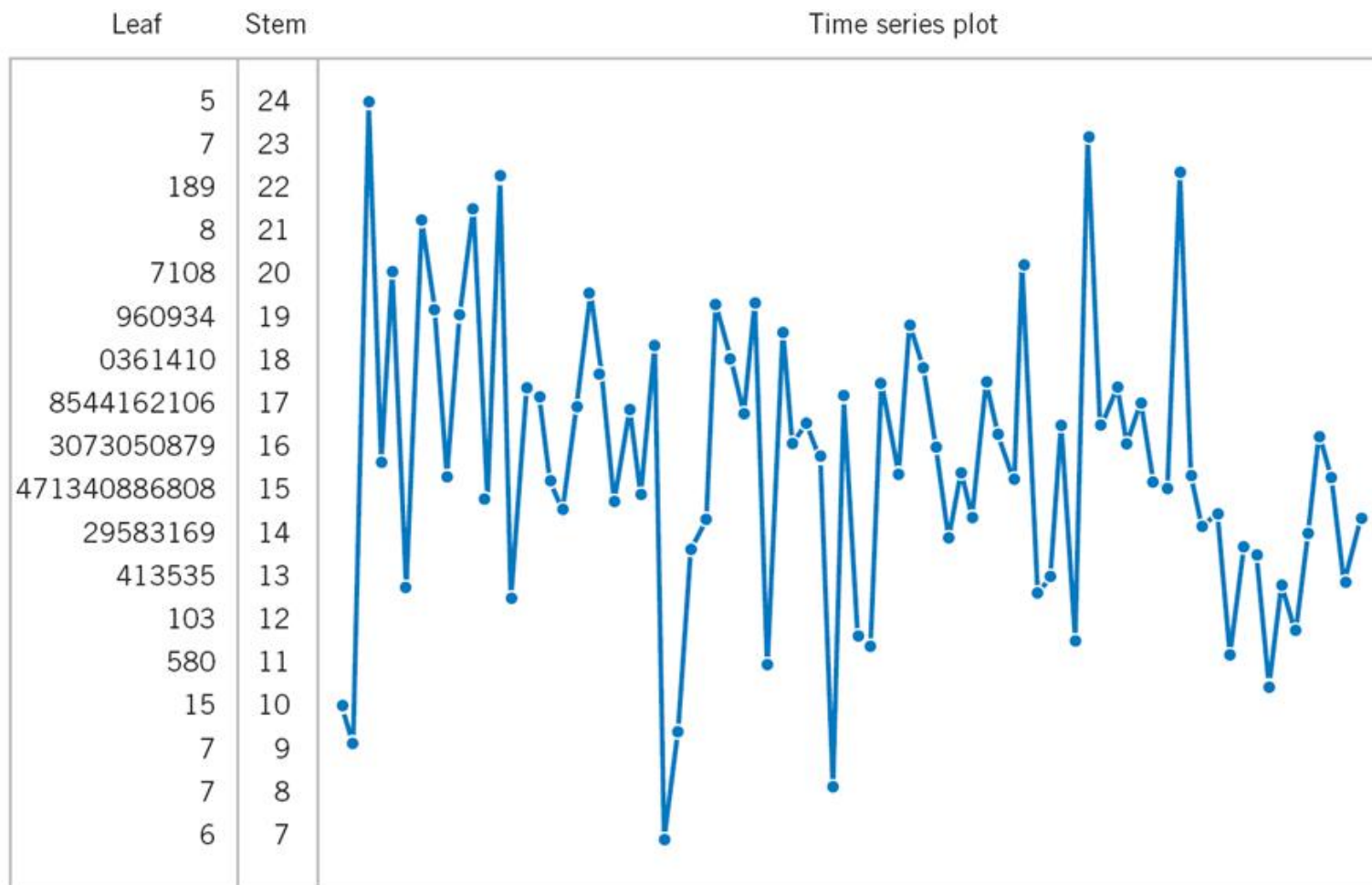
Upward trend



A cyclic variability



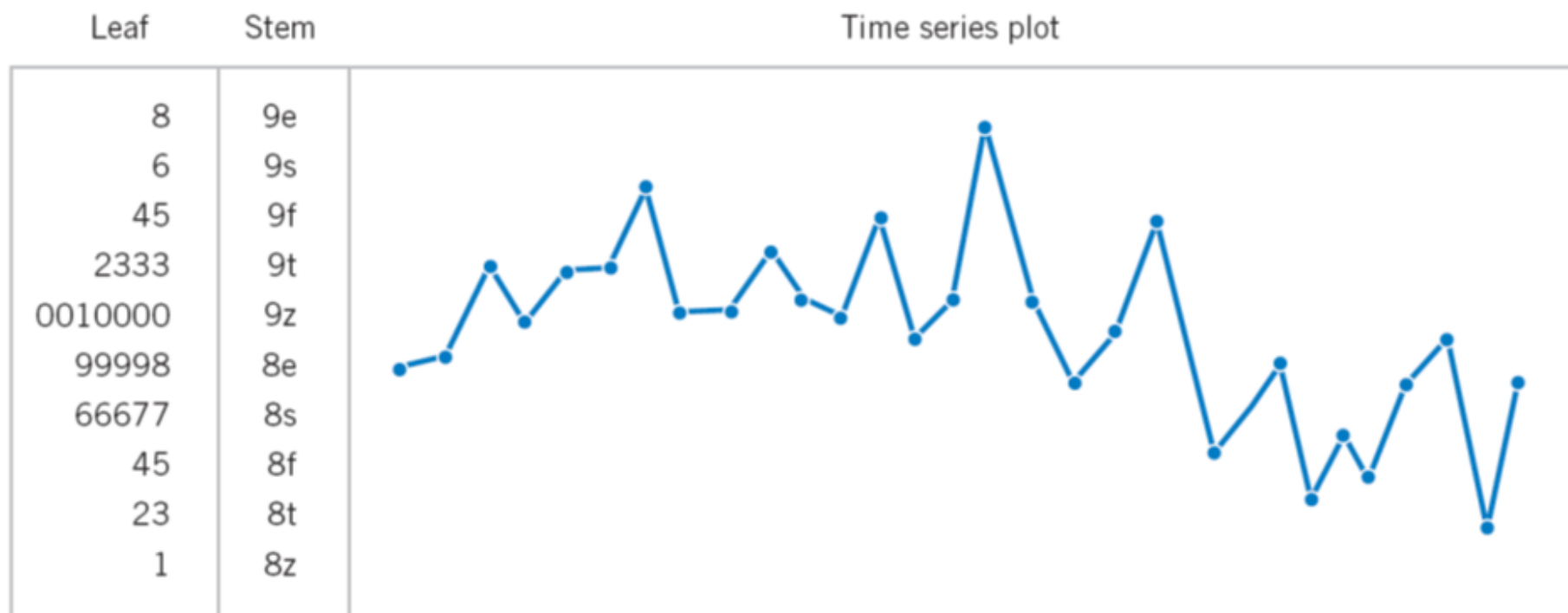
Time Series Plots (Cont.)



A digitdot plot of compressive strength data.



Time Series Plots (Cont.)



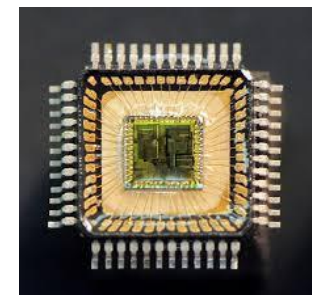
A digitdot plot of chemical process concentration readings,
observed hourly.



Multivariate Data

- ♣ The dot diagram, stem-and-leaf diagram, histogram, and box plot are descriptive displays for **univariate data**; that is, they convey descriptive information about a single variable.
- ♣ Many engineering problems involve collecting and analyzing **multivariate data**, or data on several different variables.
- ♣ In engineering studies involving multivariate data, often the objective is **to determine the relationships among the variables** or **to build an empirical model**.

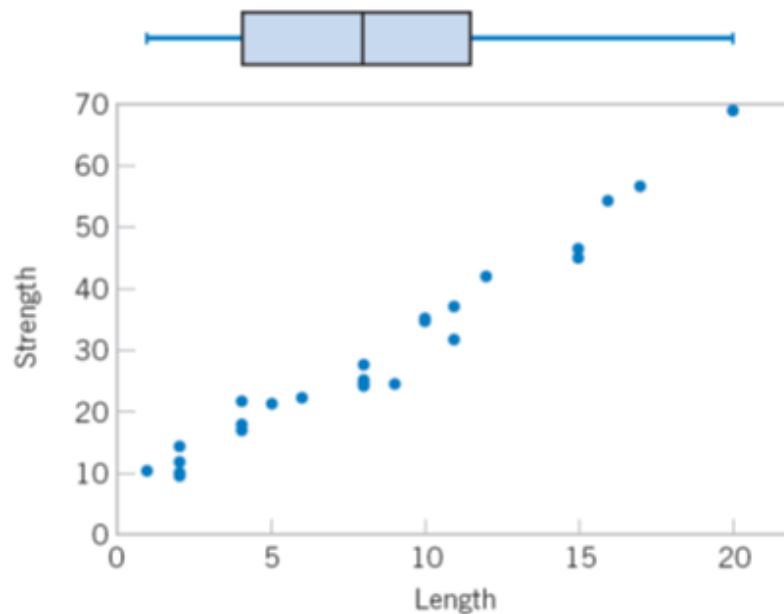
Multivariate Data (Cont.)



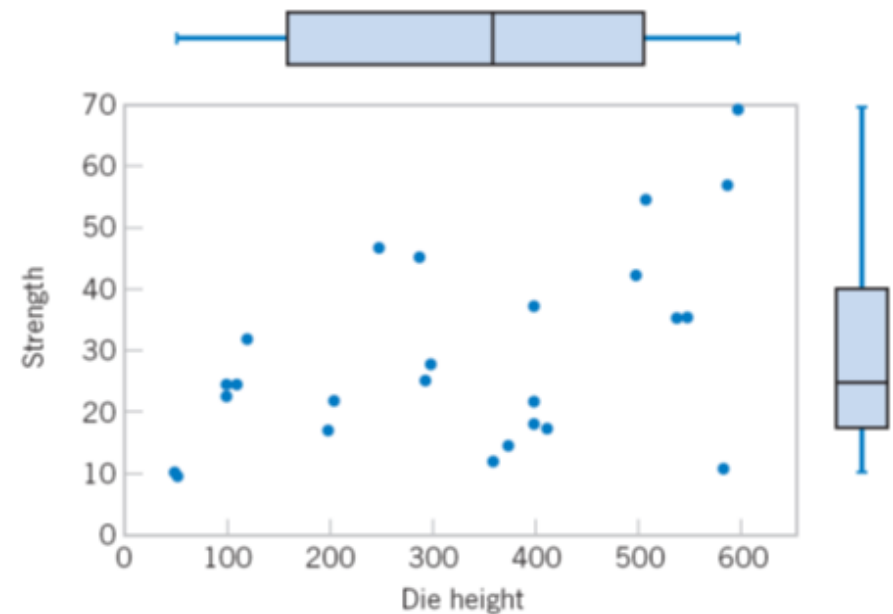
Wire Bond Data: Semiconductor Manufacturing Plant

Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2	Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2
1	9.95	2	50	14	11.66	2	360
2	24.45	8	110	15	21.65	4	205
3	31.75	11	120	16	17.89	4	400
4	35.00	10	550	17	69.00	20	600
5	25.02	8	295	18	10.30	1	585
6	16.86	4	200	19	34.93	10	540
7	14.38	2	375	20	46.59	15	250
8	9.60	2	52	21	44.88	15	290
9	24.35	9	100	22	54.12	16	510
10	27.50	8	300	23	56.63	17	590
11	17.08	4	412	24	22.13	6	100
12	37.00	11	400	25	21.15	5	400
13	41.95	12	500				

Multivariate Data (Cont.)



(a) Pull strength versus length



(b) Pull strength versus die height

Scatter diagrams and box plots for the wire bond pull strength data.



Multivariate Data (Cont.)

♣ Sample Correlation Coefficient

Given n pairs of data $(y_1, x_1), (y_2, x_2), \dots, (y_n, x_n)$, the **sample correlation coefficient** r is defined by

$$r = \frac{S_{xy}}{\sqrt{\left(\sum_{i=1}^n (x_i - \bar{x})^2\right)\left(\sum_{i=1}^n (y_i - \bar{y})^2\right)}} = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

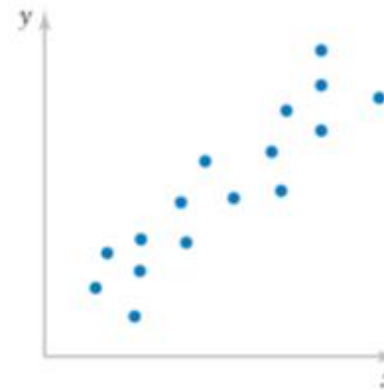
with $-1 \leq r \leq +1$.

$$\begin{aligned} S_{xy} &= \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \\ &= \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right) / n \end{aligned}$$

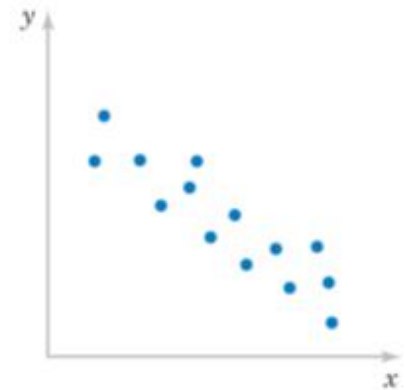
The Corrected Sum of Cross-Products: *Inner product!*

Multivariate Data (Cont.)

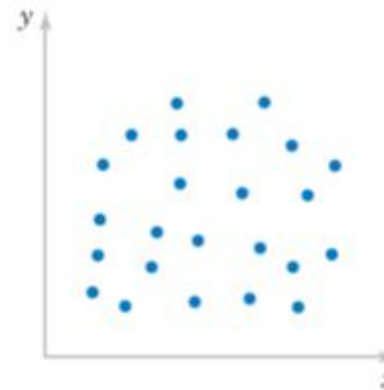
- ♣ Two variables are
 - strong if $0.8 \leq r \leq 1$,
 - moderate if $0.5 < r < 0.8$, and
 - weak if $0 \leq r \leq 0.5$.



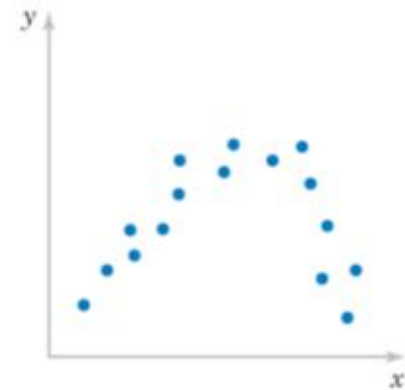
(a) r is near +1



(b) r is near -1



(c) r is near 0, y and x are unrelated



(d) r is near 0, y and x are nonlinearly related

Scatter diagrams for different values of the sample correlation coefficient r .

Data on Shampoo



Foam	Scent	Color	Residue	Region	Quality
6.3	5.3	4.8	3.1	1	91
4.4	4.9	3.5	3.9	1	87
3.9	5.3	4.8	4.7	1	82
5.1	4.2	3.1	3.6	1	83
5.6	5.1	5.5	5.1	1	83
4.6	4.7	5.1	4.1	1	84
4.8	4.8	4.8	3.3	1	90
6.5	4.5	4.3	5.2	1	84
8.7	4.3	3.9	2.9	1	97
8.3	3.9	4.7	3.9	1	93
5.1	4.3	4.5	3.6	1	82
3.3	5.4	4.3	3.6	1	84
5.9	5.7	7.2	4.1	2	87
7.7	6.6	6.7	5.6	2	80
7.1	4.4	5.8	4.1	2	84
5.5	5.6	5.6	4.4	2	84
6.3	5.4	4.8	4.6	2	82
4.3	5.5	5.5	4.1	2	79
4.6	4.1	4.3	3.1	2	81
3.4	5.0	3.4	3.4	2	83
6.4	5.4	6.6	4.8	2	81
5.5	5.3	5.3	3.8	2	84
4.7	4.1	5.0	3.7	2	83
4.1	4.0	4.1	4.0	2	80

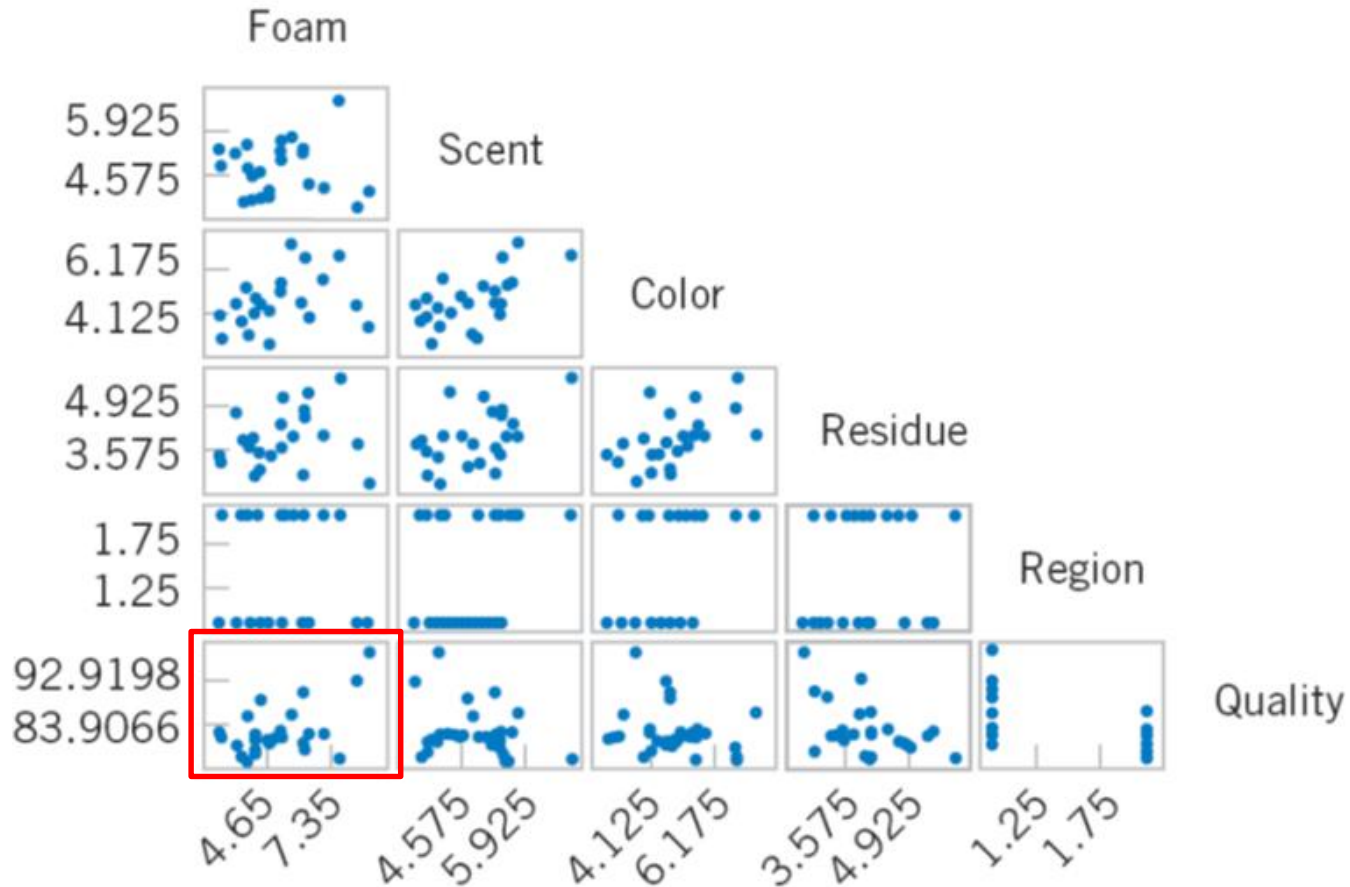


Data on Shampoo (Cont.)

	Foam	Scent	Color	Residue	Region
Scent	0.002				
Color	0.328	0.599			
Residue	0.193	0.500	0.524		
Region	-0.032	0.278	0.458	0.165	
Quality	0.512	-0.252	-0.194	-0.489	-0.507

Pairwise correlations

Data on Shampoo (Cont.)

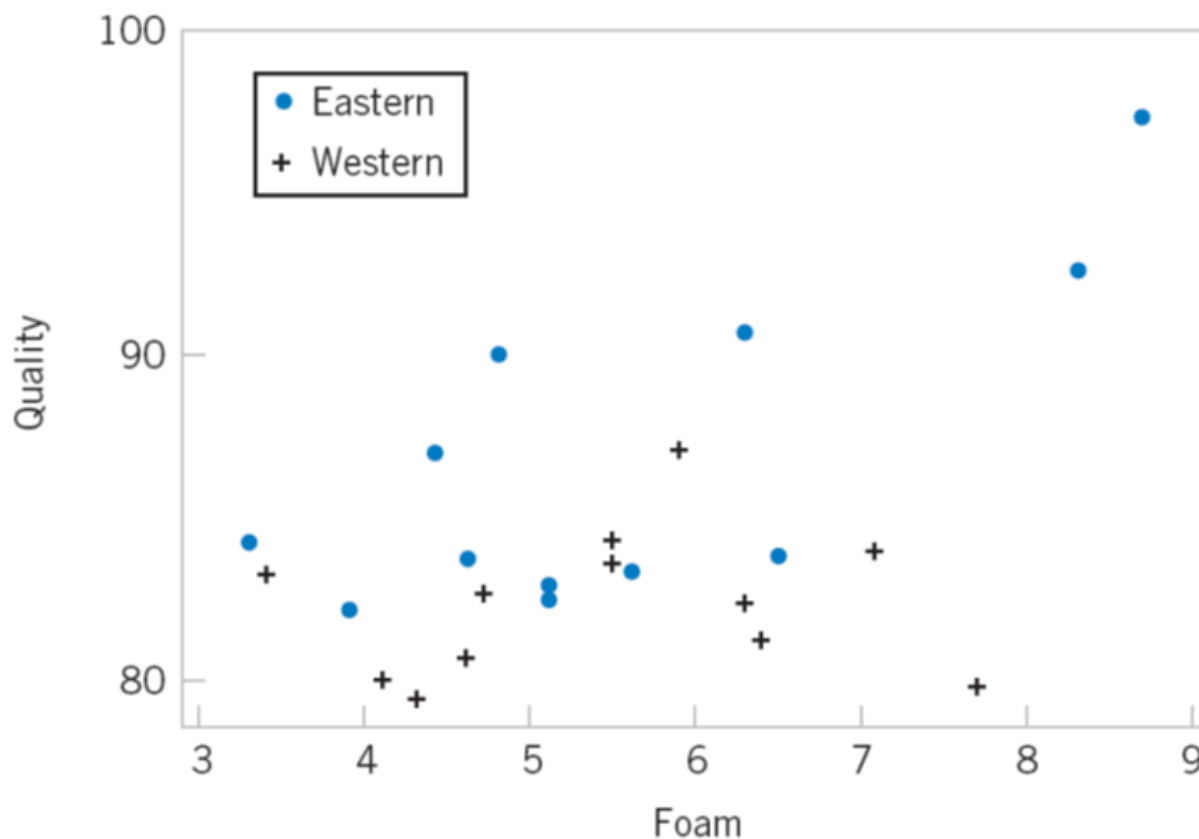


Matrix of scatter plots for the shampoo data.



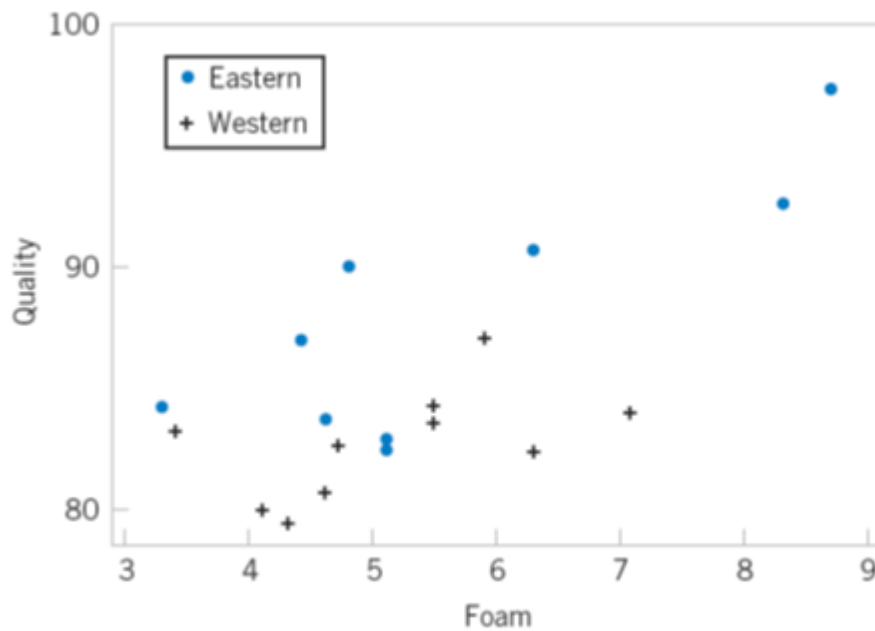
Data on Shampoo (Cont.)

Interaction between foam and region

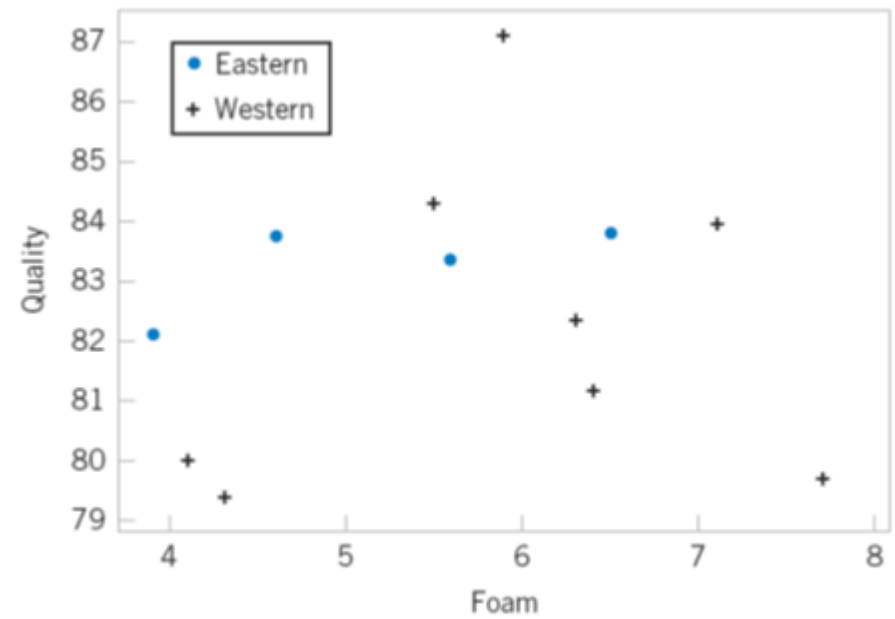


Scatter diagram of shampoo quality versus foam.

Data on Shampoo (Cont.)



(a) Residue is ≤ 4.6

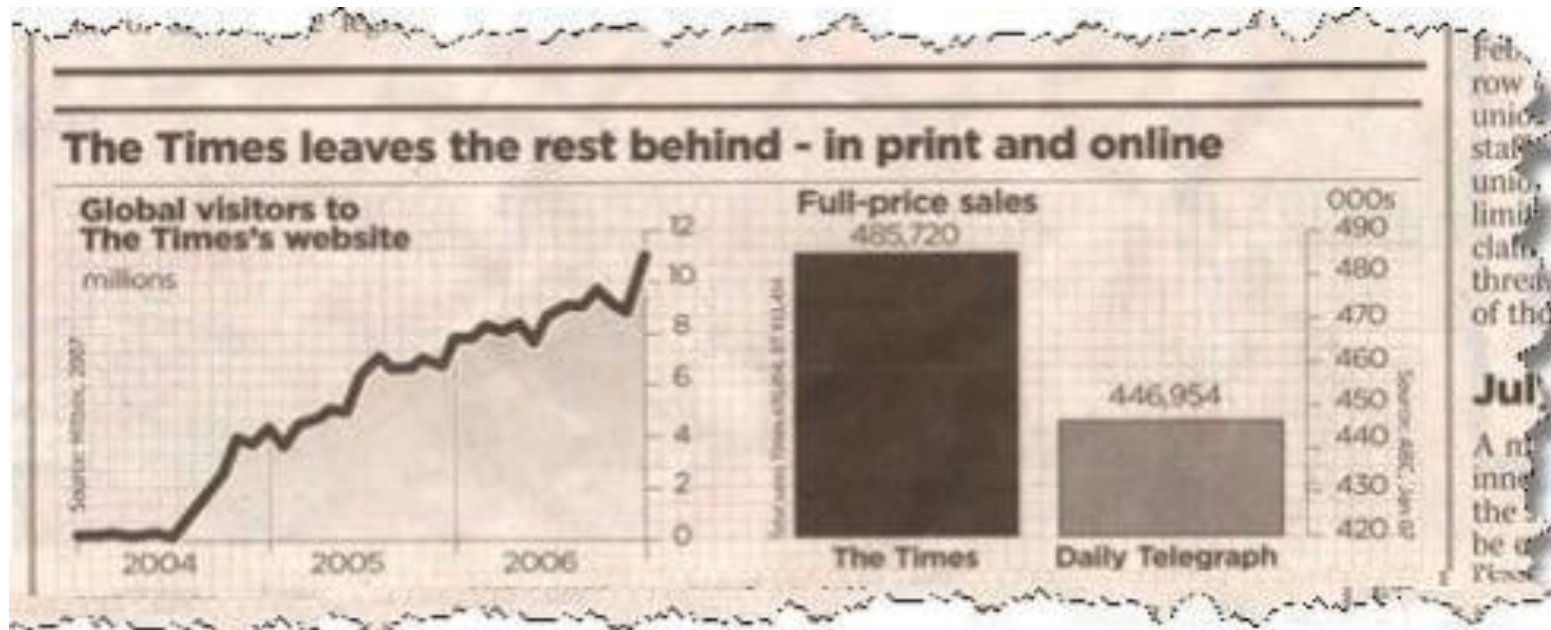


(b) Residue is ≥ 4

A coplot for the shampoo data.

Misleading Graphs

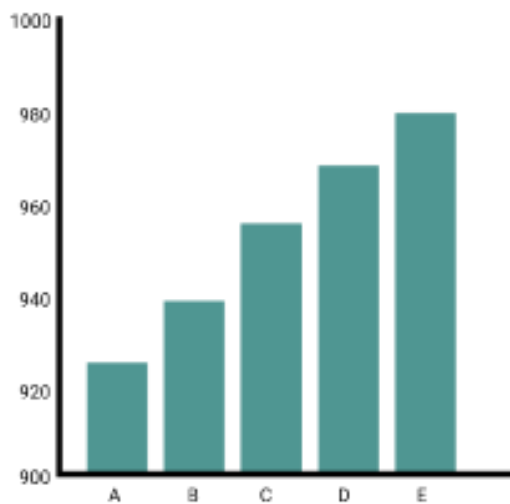
♣ 1. Manipulating the Y-axis



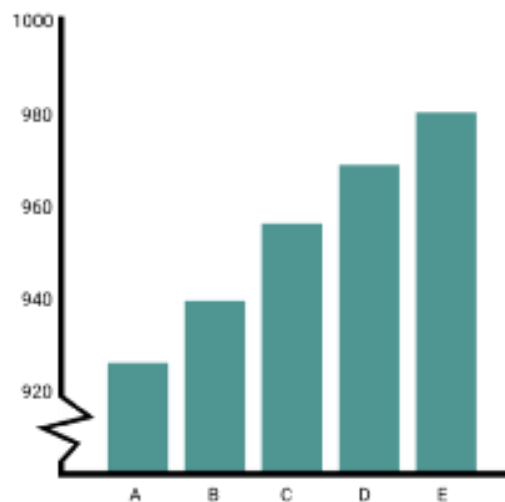
You might be thinking that the graph on the right shows The Times makes double the sales of The Daily Telegraph. But take a closer look at the scale and you'll see although The Times does make more sales, it's only beating the competition by about 10%.



Misleading Graph

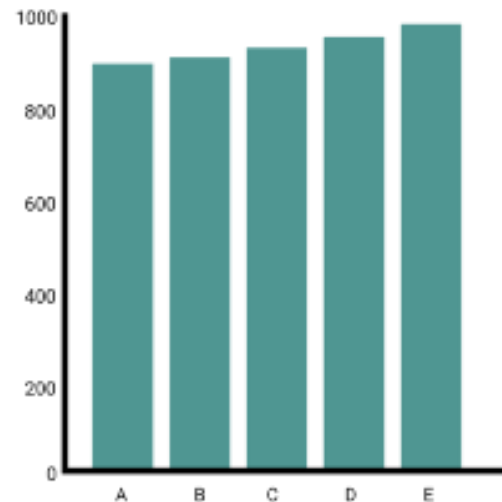


Adding Zero-Break



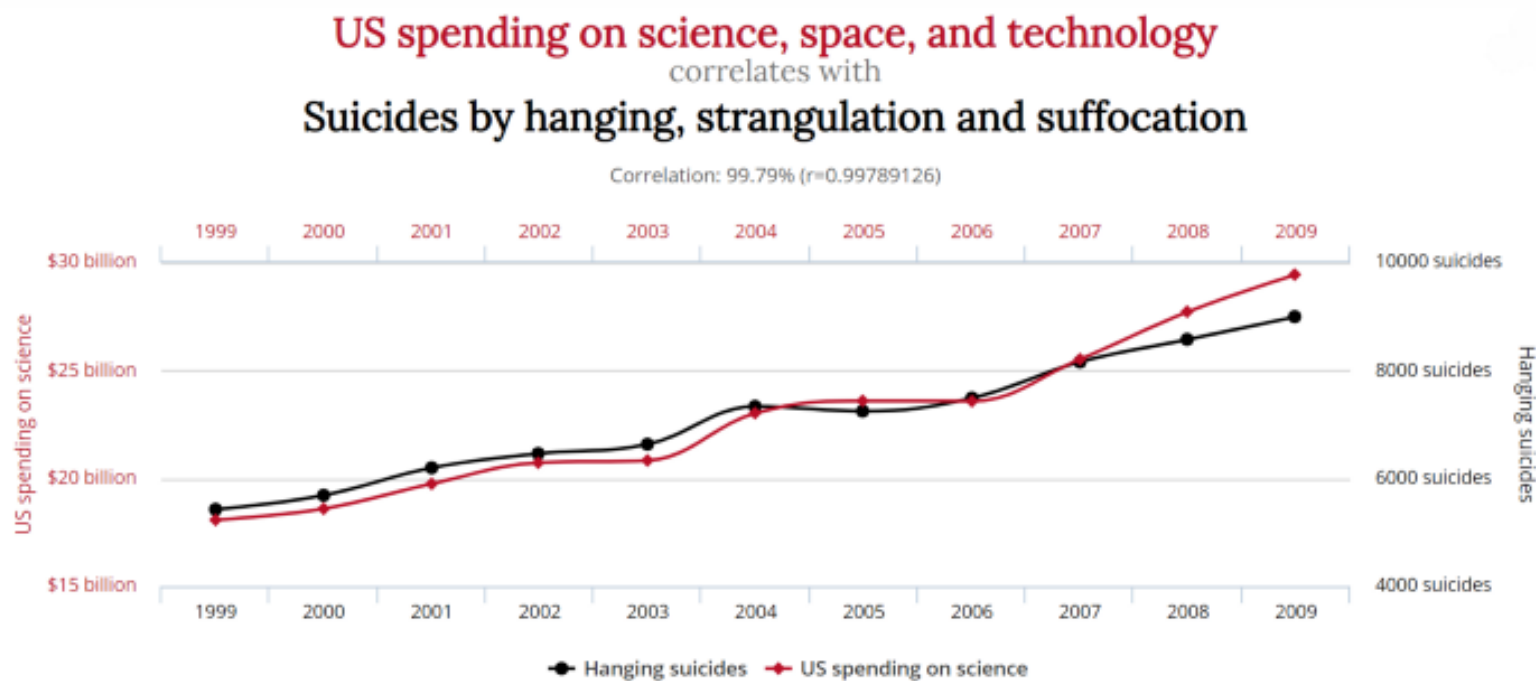
or

Starting from Zero



Misleading Graphs

♣ 2. Two Y-Axes



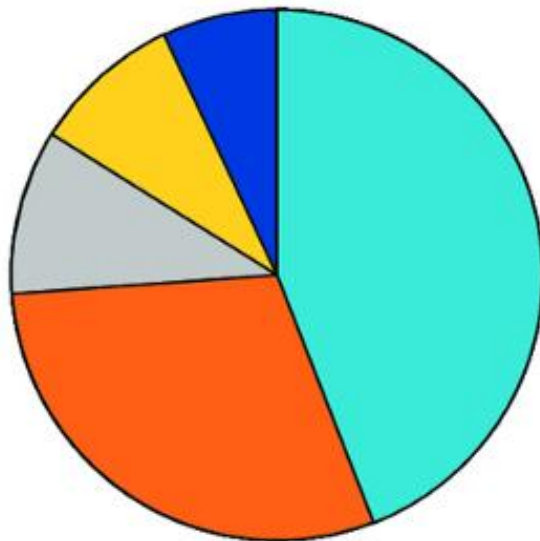


Misleading Graphs

♣ 3. 3D Graphs

A

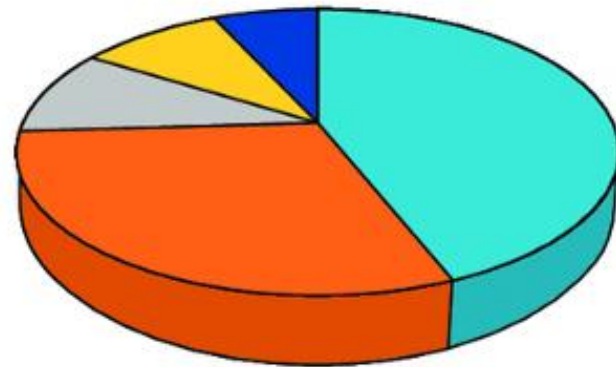
Cause of death



A B C D E

B

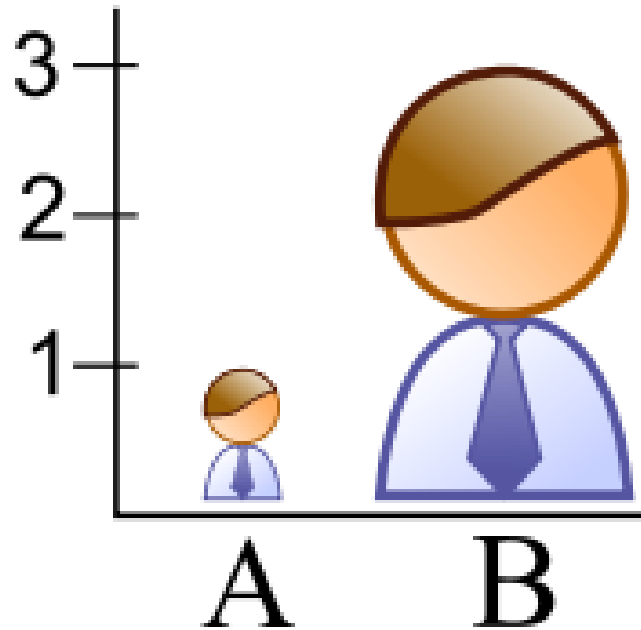
Cause of death



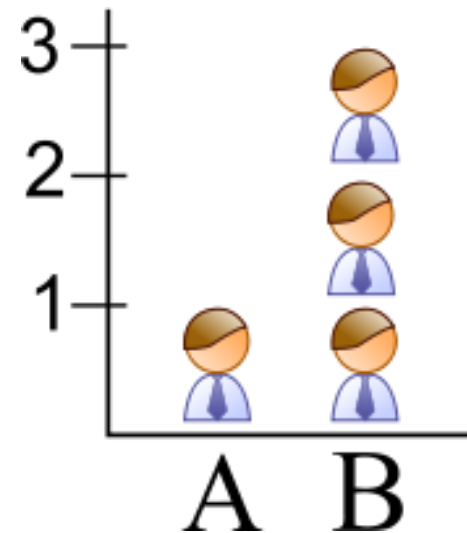
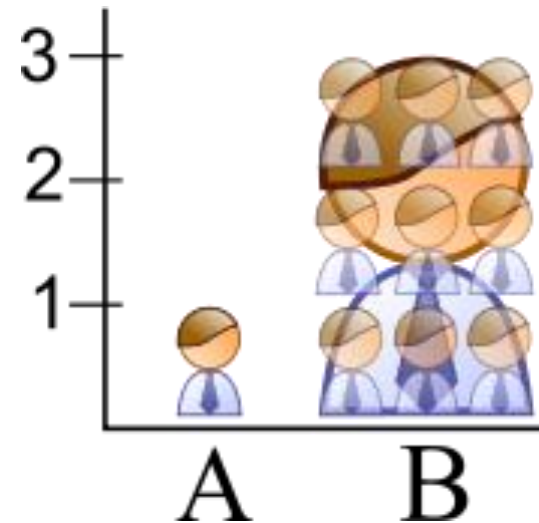
A B C D E

Misleading Graphs

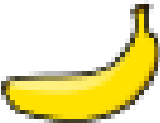
♣ 4. Improper scaling

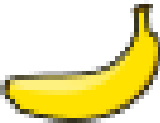
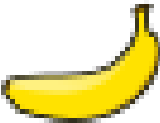


How are they difference?





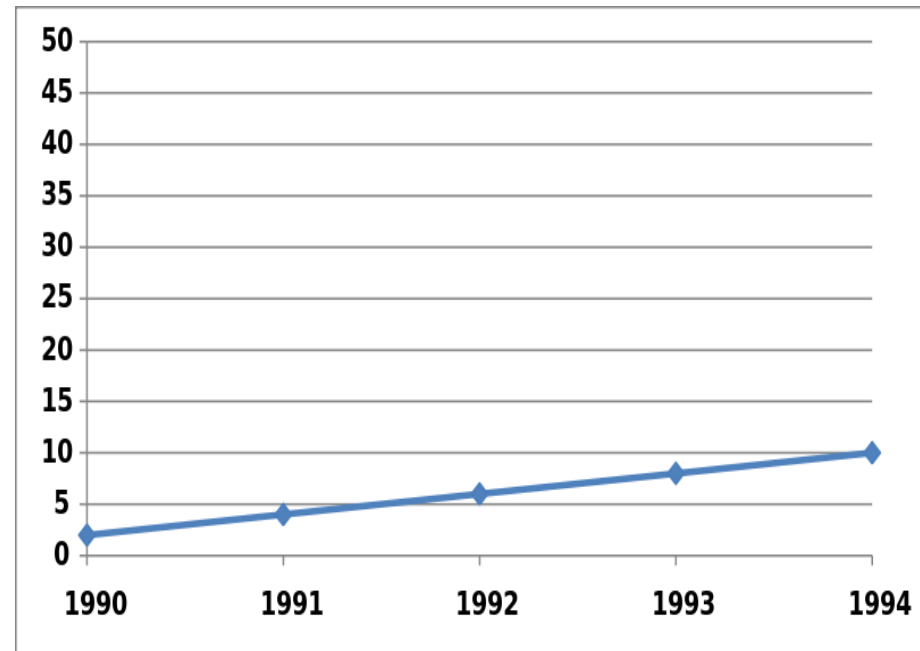
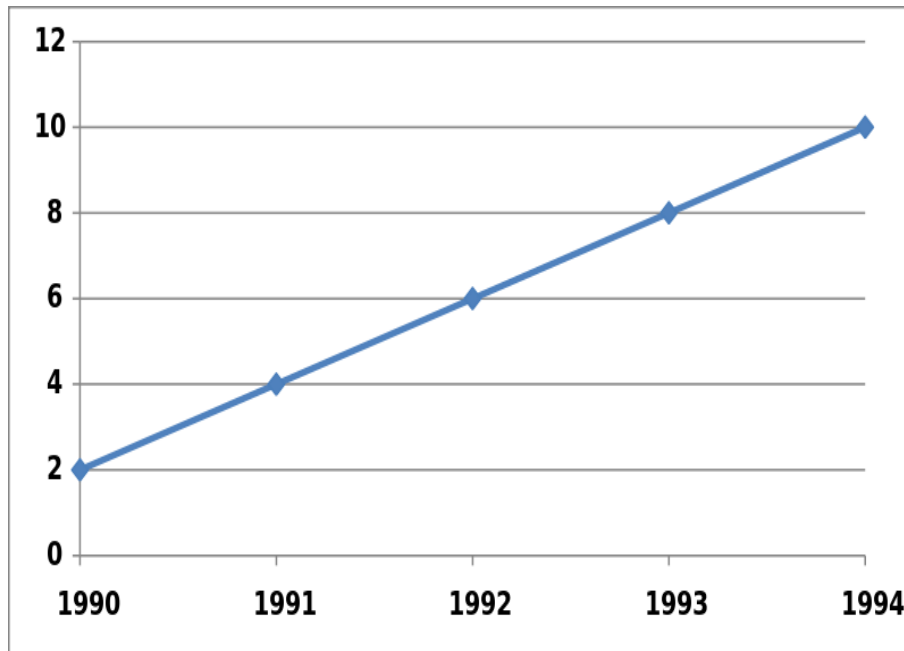
Fruit Collected	
Banana	   
Apple	   
Cherry	   

Fruit Collected	
Banana	   
Apple	   
Cherry	   



Misleading Graphs

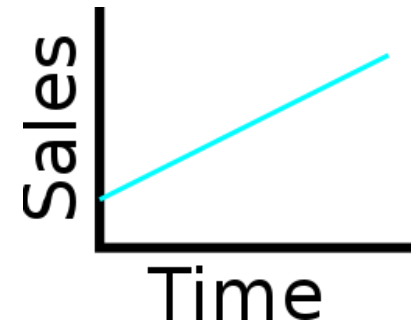
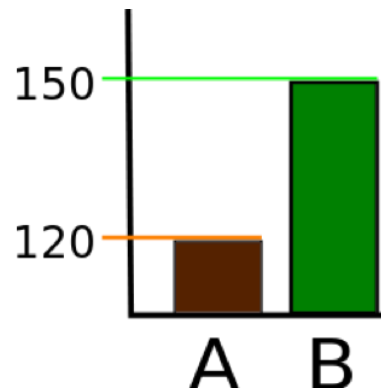
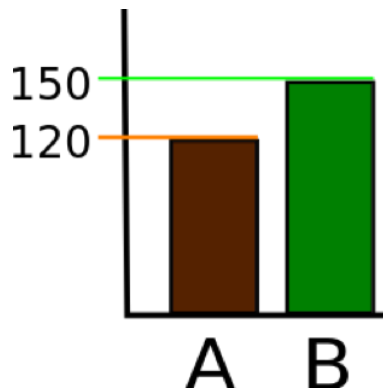
♣ 5. Truncated Graph / Torn Graph





Misleading Graphs

♣ 6. No Scale



Misleading Graphs

